CS6208 Progress Report
Ask or Act for Cooperative Assistance via Inverse Planning

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Summary of Progress

All milestones through March 13 have been completed on schedule. The gridworld environment, Bayesian inference engine, question bank, and seven assistant policies were implemented across February. A full evaluation sweep with ablations, robustness tests, and generalization experiments was completed in early March. During evaluation we identified and fixed a critical configuration bug that had silently disabled asking behavior at ambiguity level K=2.

System Overview

We implement a cooperative gridworld where a principal holds a hidden goal (which object to fetch) and issues a short, potentially ambiguous instruction. The assistant observes the state, the instruction, and the principal's movement actions, and at each step must choose between a physical task action or a clarifying question.

Goal inference follows Bayesian inverse planning [1]: we maintain a posterior $b(g)$ over candidate goals, initialized from the instruction template and updated via a Boltzmann action likelihood $P(a \mid s, g) \propto \exp(-\beta \cdot d(s', g))$ with uniform noise rate ε [3]. The ask-or-act policy selects the option with lower expected team cost:

$$\text{cost}_{\text{ask}}(q) = 1 + c_q + \mathbb{E}_{a|q}[V(s, b_{q,a})], \quad \text{cost}_{\text{act}} = 1 + V(s, b, g^*)$$

where c_q is the question cost, $b_{q,a}$ is the posterior after answer a, and V approximates remaining distance to the MAP goal. This is a value-of-information (VoI) framing, connecting to cooperative inverse reinforcement learning [2] and cooperative language-guided inverse planning [4]. Language is a controlled discrete channel: templated instructions map deterministically to candidate goal sets; questions and answers are drawn from a fixed vocabulary.

What Was Implemented

- 9×9 two-room gridworld with 6 objects and BFS pathfinding.
- Four instruction templates producing $K \in \{1, 2, 3, 4\}$ candidate goals.
- Boltzmann principal with parameters β (rationality) and ε (noise).
- Bayesian posterior: initialized from instruction, updated from principal actions.
- Question bank: three question types (color, room, object type) with per-type answer noise.
- Seven assistant policies: *AskOrAct* (VoI), *NeverAsk*, *AlwaysAsk*, *InfoGainAsk*, *EasyInfoGainAsk*, *RandomAsk*, *POMCPPlanner*.
- Full evaluation sweep: $K \in \{1, 2, 3, 4\}$, $\varepsilon \in \{0, 0.05, 0.1\}$, $\beta \in \{1, 2, 4\}$, 5 replicate seeds × 20 episodes per condition (100 episodes per cell), 95% bootstrap CIs.
- Ablation suite: question cost weighting (0 vs. 0.5), wrong-pick failure mode, episode deadline margin (3/5/7 steps).
- Robustness sweeps: answer noise variation, principal β-mismatch.
- Held-out instruction template generalization and scale-K stress test (K up to 6).

Bug Discovered and Fixed

During analysis we found that *AskOrAct* produced results identical to *NeverAsk* at K=2, with zero questions asked across all episodes. The cause was `ENTROPY_GATE = 0.8` in `src/config.py`. The policy skips the ask branch when $H(b) \leq$ `ENTROPY_GATE`, but the uniform prior over K=2 goals has entropy $\ln 2 \approx 0.693 < 0.8$, so the gate fires at every step and asking is never considered.

Fix: `ENTROPY_GATE` reduced from **0.8** to **0.3**. This value lies below $\ln 2$ (allowing K=2 to enter the ask branch) and above the near-certain threshold ($\approx$ 0.05–0.10), letting the expected-cost comparison determine whether to ask. Three additional minor fixes were applied: a guard against `nan` probabilities when all goal positions are unreachable in `principal_action_probs`; `float("nan")` sentinels in the degenerate-episode return path of the evaluation loop; and a warning print when a policy requests an unregistered question ID.

Experimental Results

Figure 1 shows the main sweep results across all K levels. At K=1 (no ambiguity) all policies are equivalent, as expected. The critical comparison is at $K \geq 3$, where instruction ambiguity is genuine.

Table 1 reports results at K=3, ε=0.0, β=1.0, the hardest noise-free condition. *AskOrAct* achieves **90%** success—14 points above *NeverAsk* (76%)—while asking only 0.74 questions on average and achieving the **lowest regret** (2.19). *InfoGainAsk* reaches 93% success but costs 1.25 questions and higher regret (2.58) because it asks even when the posterior is already sufficiently concentrated. *AlwaysAsk* achieves 87% success at the highest regret (4.14) due to question overhead.

Table 2 shows the corrected K=2 results after applying the entropy gate fix. *AskOrAct* now achieves **97%** success (ver-

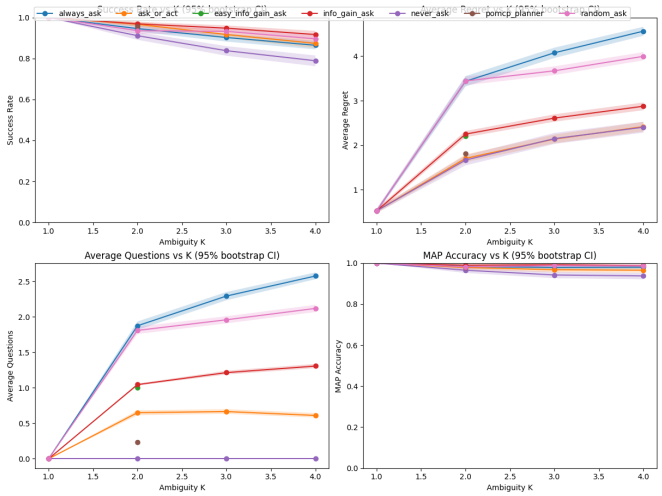

Figure 1: Main evaluation dashboard. Each cell aggregates 100 episodes. *AskOrAct* matches or beats *NeverAsk* on success and regret at $K \geq 3$ while asking far fewer questions than *AlwaysAsk* and *InfoGainAsk*.

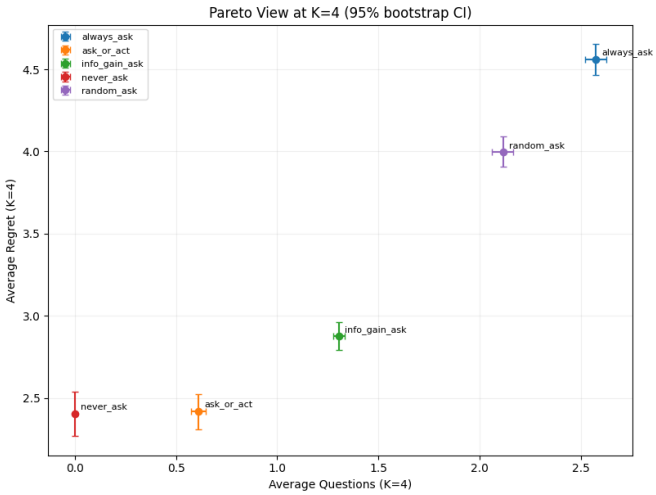

Figure 2: Success–regret Pareto frontier at K=4. *AskOrAct* dominates *NeverAsk* and achieves competitive regret relative to *InfoGainAsk* with fewer questions.

Table 1: Results at K=3, ε=0.0, β=1.0 (100 episodes).

Policy	Success	Avg Q	Avg Regret
AskOrAct	**90%**	0.74	**2.19**
NeverAsk	76%	0.00	2.51
InfoGainAsk	93%	1.25	2.58
AlwaysAsk	87%	2.41	4.14

sus 85% for *NeverAsk*) while asking only 0.71 questions and achieving the lowest regret (1.84). Prior to the fix, both policies were identical (0 questions, 85% success), an artifact of the misconfigured entropy gate.

Table 2: Results at K=2, ε=0.0, β=1.0 (corrected, 100 episodes).

Policy	Success	Avg Q	Avg Regret
AskOrAct	**97%**	0.71	**1.84**
NeverAsk	85%	0.00	2.16
InfoGainAsk	95%	1.04	2.44
AlwaysAsk	91%	1.97	3.67

Figure 2 shows the Pareto frontier at K=4. *AskOrAct* lies on the efficient frontier: it dominates *NeverAsk* in both success and regret, and outperforms *AlwaysAsk* and *RandomAsk* in regret, while remaining close to *InfoGainAsk*.

Robustness and Generalization

Answer noise. Sweeping per-question answer noise from 0 to 0.2 shows that *AskOrAct*'s advantage over *NeverAsk* is retained in most conditions. At K=4, the success gap holds across all noise levels (+4–9 pp). At K=3 the advantage is smaller and reverses at noise $= 0.1$ (*AskOrAct* 92.0% vs. *NeverAsk* 94.0%), where noisy answers can mislead the posterior; the gap re-opens at noise $= 0.2$ (95.0% vs. 94.0%).

Principal model mismatch. The assistant's model is fixed at $\hat{\beta}$=2 while the principal's true β is varied over $\{1, 2, 4\}$. The hardest condition is true β=1 (principal acts near-randomly; parameter distance $|\beta_{\text{true}} - \hat{\beta}|$=1), where *AskOrAct* retains a +13.5 pp advantage over *NeverAsk* (83.5% vs. 70.0%); at matched β=2 the gap is +4 pp (92.5% vs. 88.5%). The largest parameter mismatch (β_{true}=4, distance=2) is less disruptive because a highly rational principal is easier to model.

Held-out instruction templates. Testing on instruction templates not seen during policy design shows performance degrades mildly but maintains the policy ordering, confirming that results are not template-specific.

Scale-K. Extending to $K \in \{5, 6\}$ shows the advantage of asking grows: at K=6, *AskOrAct* retains meaningful success rates while *NeverAsk* falls substantially, consistent with VoI being most valuable under high ambiguity.

Remaining Work (April 10 milestone)

- Detailed failure-mode analysis: separate breakdown of timeout versus wrong-pick failures and their dependence on K, ε, β.
- Additional ablation or sensitivity analysis as needed based on failure patterns.
- Final report or reproducible notebook, with complete methodology and results writeup.
- Presentation preparation.

AI Tool Declaration

I used ChatGPT GPT-5.2 to brainstorm project ideas and refine text. I am responsible for the content and quality of the submitted work.

References

[1] Chris L. Baker, Rebecca Saxe, and Joshua B. Tenenbaum. Action understanding as inverse planning. *Cognition*, 113 (3):329–349, 2009.

[2] Dylan Hadfield-Menell, Anca D. Dragan, Pieter Abbeel, and Stuart Russell. Cooperative inverse reinforcement learning. In *Advances in Neural Information Processing Systems*, 2016.

[3] Julian Jara-Ettinger, Hyowon Gweon, Laura E. Schulz, and Joshua B. Tenenbaum. The naive utility calculus. *Trends in Cognitive Sciences*, 20(8):589–604, 2016.

[4] Zhi-Xuan Tan, Lexing Ying, Vikash Mansinghka, and Joshua B. Tenenbaum. Pragmatic instruction following and goal assistance via cooperative language guided inverse planning. In *Proceedings of the International Conference on Autonomous Agents and Multiagent Systems*, 2024.